Highlights

Cost-aware collaboration framework of vision–language experts from benchmark with different granularities

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- **Framework:** Bench-CoE builds cost-aware routers from benchmark records, label-free.
- **Three granularities:** query-, subject-, and state-level variants of one routing form.
- **Theory:** Beta-posterior shrinkage; few states suit shift; a label-free test sets K.
- **Experiments:** three variants and ten published routers on VL-RouterBench, ID and OOD.
- **Conclusion:** per-query labels overfit; two cost-aware learned states transfer best.

Cost-aware collaboration framework of vision–language experts from benchmark with different granularities

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ABSTRACT

As vision–language expert models continue to advance, the tasks they handle have become increasingly complex and diverse, making it difficult for any single model to achieve optimal performance across all tasks cost-effectively. Expert collaboration integrates the strengths of diverse models while controlling expenditures by routing each query to the corresponding expert in a vision–language model pool. However, three primary challenges remain: high routing supervision costs from running every expert on all training queries to obtain labels, ambiguous supervision granularity regarding whether labels should be applied per query or aggregated into coarser groups, and expert selection based purely on accuracy, which overlooks price disparities spanning up to multiple orders of magnitude. To address these challenges, we propose Bench-CoE, a framework under which variants across different granularities are formulated to suit distinct scenarios. To eliminate data collection and annotation costs, Bench-CoE reuses published results from comprehensive benchmarks that already evaluate each expert's per-sample performance. To resolve granularity ambiguity, it models each router as a combination of state assignment and state-to-expert mapping, deriving three variants differing solely in granularity. Furthermore, each variant explicitly balances expert cost against competence rather than relying solely on accuracy. Evaluated on VL-RouterBench, our results demonstrate that while the cost-aware query-level variant improves its seen-dataset trade-off, its Rank Score drops by about one-seventh on out-of-distribution datasets, whereas two learned states achieve the best transfer performance at one-fourth the cost of the strongest expert, showing that granularity and cost awareness ultimately determine how routers transfer.

1. Introduction

Large language models (LLMs) are capable of performing various natural language processing tasks through auto-regressive prediction conditioned on task prompts [1]. The ability of LLMs to describe and unify tasks makes them key components in current visual understanding tasks, leading to the emergence of large multimodal models (LMMs) [2, 3]. While these LLM- and LMM-driven experts can handle a wide range of visual and language tasks, they possess different areas of expertise and exhibit significant performance variations across tasks. As the capabilities of experts continue to improve, benchmark evaluations have also become increasingly complex and diverse [4–6]. For instance, MMLU-Pro [6] assesses multi-subject reasoning in language tasks, while MMMU [5] evaluates cross-domain reasoning in multimodal tasks. It is difficult for a single model to achieve optimal performance across all tasks

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at low cost, and attaining comparable effectiveness often requires a larger model and a higher inference cost, which strains computational resources and practical applications.

To integrate the strengths of different models while controlling cost, researchers have proposed two paradigms. Mixture of experts [7–9] introduces multiple expert sub-networks and activates only a subset of them for each token, which reduces computation. Its experts, however, are sub-modules of one model, so their functionality and the decision process behind a task are hard to understand, and they cannot be evaluated or replaced through external benchmarks. Collaboration of experts [10–13] instead integrates independently deployable models by routing each query to a specific expert through a router. The routers packaged with VL-RouterBench [4] follow this design, among them kNN routing from benchmark records [14] and the dual contrastive router RouterDC [15], and RouteLLM [12] and Hybrid LLM [16] add an explicit trade-off between quality and cost. These routers outperform the strongest fixed expert on the datasets they were built from, but they rely on routing labels collected for the purpose, and in the benchmark's out-of-distribution (OOD) setting most of them lose much of their advantage [4]. Coarser supervision has been explored as an alternative. UniRoute [17] profiles each model on clusters of representative prompts. On VL-RouterBench [4], an oracle that always picks the cheapest correct expert is both far more accurate and several times cheaper than the strongest single model. Is there a way to keep a router accurate and economical beyond its source data without the cost of collecting labels?

Three challenges stand in the way. **Costly routing supervision.** A router needs to know which expert answers which query correctly, and collecting this means running every candidate on every training query or asking people to compare outputs [12, 13, 16]. The cost grows with the number of queries times the number of experts, and for API models it is paid in money as well as compute. It cannot simply be avoided by labelling fewer queries, because a router trained on a thin sample of the input space has no evidence where deployment data lie [4, 14]. **Unknown supervision granularity.** The same records can be aggregated per query or over coarser groups such as benchmark subjects and clusters of inputs, and each existing router fixes one of these choices by construction [4, 17]. The choice is hard because it is a bias–variance trade-off. A fine partition describes each region precisely but estimates its competence from few records, and local-accuracy selection is known to be sensitive to exactly this [18, 19]. A coarse partition is stable but may merge regions where different experts are best. **Accuracy-only expert selection.** Most routers, including subject-to-expert maps read from leaderboards, choose the expert that is most likely to be correct and ignore what it costs [10, 13]. In a pool whose prices span multiple orders of magnitude, a small gain in accuracy can buy an expert several times more expensive, and every chance error in a competence estimate can do the same. Adding cost is hard because it can enter at several places, in the labels, in the map or as a budget, and because a trade-off tuned on the source data need not hold on new data [4, 16, 20].

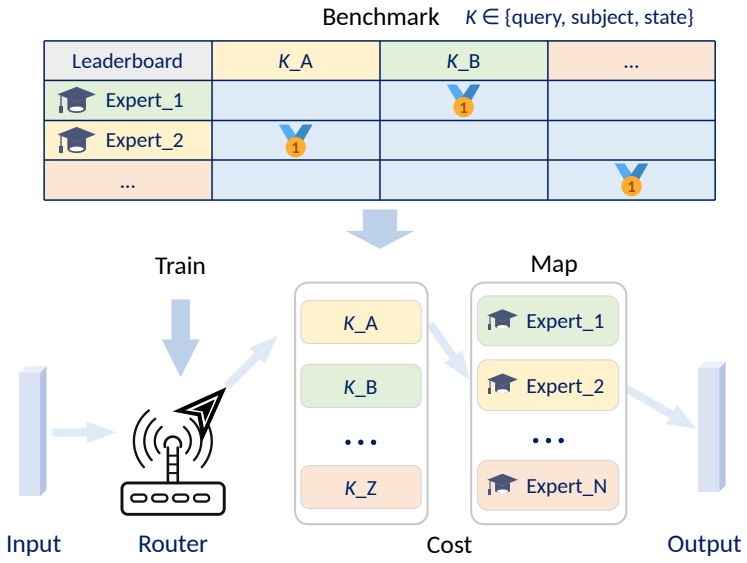


Figure 1: The framework of Bench-CoE. It consists of a set of specialized expert models, a router for expert assignment, and a benchmark dataset for training the router. $K \in \{\text{query}, \text{subject}, \text{state}\}$ is the granularity at which queries and benchmark records are grouped, and its three values give Query-, Subject- and State-Bench-CoE.

The evaluation results that benchmarks already publish point to an answer, because the same benchmark holds the evidence needed to learn the routing choice: it publishes the scored output and the cost of every expert on every sample. We propose Bench-CoE, a framework that builds routers from benchmark results alone and weighs cost alongside accuracy, as shown in Fig. 1. Bench-CoE consists of a set of expert models, a router that assigns each query to one expert, and an integrated benchmark whose published results supply the supervision for training the router and the test sets for evaluating it. Every router is written as $r(x) = h(z(x))$, an assignment z of the query to one of K states followed by a fixed map h from states to experts.

To address the challenge of costly routing supervision, Bench-CoE builds every router from benchmark records alone, since a benchmark that scores many models on many samples already publishes a correctness label and an invocation cost for every sample and expert, so training needs no further inference or annotation. To address the challenge of unknown supervision granularity, it writes every router in one state-to-expert form and derives three variants of decreasing granularity. Query-Bench-CoE gives each training sample its own state, Subject-Bench-CoE uses the subjects of a multi-discipline benchmark as states with a map read from the per-subject leaderboard, and State-Bench-CoE clusters frozen input representations into a few states whose expert competences are estimated by shrinkage. To address the challenge of accuracy-only expert selection, every variant carries the benchmark's own trade-off between correctness and cost [4], attached to the sample labels at the query level and applied as a penalty on pooled competence at the subject and state levels, and each variant is reported with and without it, with its single cost knob selected on source data.

To evaluate variants of different granularities under the framework, we train and evaluate each variant on VL-RouterBench [4], an integrated benchmark of 14 datasets and 17 experts, in distribution and on datasets withheld from router construction, beside the ten routers released with the benchmark. Granularity decides transfer. On 11-ID the cost-aware query-level TinyLLaVA router improves on its cost-free form, reaching a Rank Score of 73.83, above the benchmark's published RouterDC, VLC and MLP values but below two-state State-Bench-CoE at 74.09; on 3-OOD it falls to 63.32. The cost-free subject-level variant holds the highest seen-dataset accuracy among routers at several times the cost of the learned routers and collapses under shift, where its cost-aware map recovers much of the loss. State-Bench-CoE with two learned states stays close to the best router in distribution, leads on OOD datasets, and costs a fraction of the strongest expert. The cost-aware rule improves the OOD subject map, and a small learned partition adds the remaining transfer gain.

Our contributions are as follows.

1. We propose Bench-CoE, a framework that trains vision–language routers on the published results of an integrated benchmark without collecting or annotating data, weighs cost alongside accuracy, and expresses routing at three granularities as variants of one state-to-expert form.
2. An analysis of the framework. It interprets the shrinkage profile as a Beta posterior mean with a closed-form condition for lower error and explains why the number of states should be small under dataset shift. It also casts the cost term as the Lagrangian of the benchmark's own objective and yields a label-free map-stability diagnostic for choosing K.
3. A controlled study of the three variants beside ten published routers on VL-RouterBench, in the in-distribution (ID) and OOD settings. It shows that per-query supervision fits the source datasets but transfers poorly, that a cost term in the map recovers most of the transfer that a benchmark taxonomy loses, and that a few learned states transfer best.

2. Related work

2.1. From mixture of experts to collaboration of experts

Two paradigms combine expert models, and they route at different levels. Mixture of experts uses a gating network to send each input token to a sparse subset of internal expert modules, expanding model capacity without a proportional increase in computation. Jacobs et al. introduced adaptive mixtures with a learned gate [9], Shazeer et al. scaled sparsely gated mixtures to large networks [7], and Switch Transformer simplified routing to one expert per token [8]. These experts remain jointly trained internal components rather than independently deployable models, so their capabilities are hard to identify, update or validate through external benchmarks. Collaboration of experts instead treats independently deployable models as experts and assigns each query to one or more of them [10, 21]. Multi-inference

schemes obtain outputs from several models and then select, fuse or cascade them, as LLM-Blender [11], Fusion of Experts [22] and cluster-route-escalate cascades [23] do, and they buy quality with a second inference. Single-inference schemes choose one expert before any answer is generated. The first were per-query selectors trained from benchmark records, as per-expert correctness predictors [14] or through reward distillation in ZOOTER [13]. Later routers improved the selector's representation, through dual contrastive embedding of queries and models in RouterDC [15], item-response modelling of difficulty and ability [24], tags [25] and query–model graphs [26], and UniRoute [17] made unseen models routable by representing each model through its errors on clusters of representative prompts. Query-Bench-CoE abstracts these single-inference routers, and the reference routers of our experiments, packaged by VL-RouterBench [4], belong to this line.

2.2. Cost-aware routing

Routing became a trade-off between quality and cost once the experts carried different prices. RouteLLM learned a strong-versus-weak decision from preference data [12], Hybrid LLM [16] routed between a small and a large model under a quality threshold, and FrugalGPT [20] invoked progressively stronger models in a cascade until an answer was accepted. TensorOpera Router [27] framed the task as a trilemma of quality, cost and throughput. VL-RouterBench [4] made the trade-off part of router training, through soft labels that weigh each correct expert by an exponential of its cost, and part of evaluation, through a Rank Score that combines accuracy and normalized cost. In all of these the cost enters per query, either in the label or in a threshold on one decision. Bench-CoE keeps the benchmark's objective but applies it in the map of each variant, where cost acts on competence pooled over many records.

2.3. Benchmarks as supervision and as testbeds

Benchmarks play two roles in routing research, as supervision for building a router and as testbeds for comparing routers. Log-based benchmarks made routing reproducible. RouterBench [28] introduced aligned quality and cost matrices for text routing, RouterEval [29] scaled the pool to thousands of models, RouterArena [30] added a multi-metric leaderboard, and MMR-Bench [31] is a concurrent multimodal benchmark. VL-RouterBench [4] integrates 14 vision–language datasets and 17 experts and adds a protocol that withholds whole datasets from router construction, which makes transfer measurable. As supervision, LLM routing with benchmark datasets [14] trains per-model correctness predictors on benchmark queries, which still requires the outcome of every candidate on every query. Multi-discipline benchmarks such as MMLU-Pro [6] and MMMU [5] offer a cheaper signal, since their subject taxonomies and per-subject leaderboards are published with the benchmark, so a subject-to-expert map can be read from them directly. MODULAR-CoE [32] likewise decouples routing into an input classifier and a category-to-expert map, but collects separate labels for both. Bench-CoE uses one integrated benchmark in both roles at once: its published records train every variant, and its OOD setting evaluates every variant.

2.4. Dynamic classifier selection and the size of competence regions

Multiple-classifier systems moved from fusing every classifier's output to selecting one classifier per instance, on the observation that competence is local [33, 34]. Static combination ran the whole pool and fused outputs by fixed rules [35, 36], whereas dynamic selection estimated for each test instance which classifier was competent in its neighbourhood. Woods et al. [18] used local accuracy in a nearest-neighbour region, Giacinto and Roli [37] refined the region through multiple-classifier behaviour, and Didaci et al. [19] showed that the estimates are sensitive to the region's size. A second strand replaced the per-instance neighbourhood by a fixed partition: Kuncheva's clustering-and-selection model [38] clusters the feature space and stores the locally best classifier per cluster, and Liu and Yuan [39] combined clustering with selection in the same spirit. Dynamic ensemble selection generalized the choice to subsets, through k-nearest oracles [40] and overproduce-and-choose strategies [41], and later work modelled competence probabilistically [42] or by meta-learning over several competence features [43]. The most recent work acts on the pool itself, pruning base classifiers whose local competence estimates are unreliable [44, 45], generating local pools on demand [46], and selecting by prediction confidence [47]. State-Bench-CoE belongs to the clustering-and-selection strand, revisited for frozen generative experts with prices and for transfer across datasets rather than across folds.

3. The Bench-CoE framework

3.1. Supervision from benchmark records

Bench-CoE consists of a pool of expert models, a router that assigns each query to one expert, and a benchmark whose records supply the supervision for building the router (Fig. 2). Let $x = (I,t)$ be an image–text query and $\mathscr{M} = \{m_1, \ldots, m_M\}$ a fixed pool of M experts. A set of source datasets provides, for every sample x_i and every expert m, a correctness label $Y_{im} \in \{0,1\}$ from rule-based scoring of the expert's answer and an invocation cost c_{im} from logged token counts and per-token prices. When a benchmark publishes its results, these records exist already, because producing the leaderboard required running every expert on every sample and scoring every answer. Bench-CoE uses them as its only supervision and needs no further inference and no annotation. Each variant reads a different summary of the same records. The query-level router reads a label per sample, the subject-level router reads the per-subject mean of Y_{im} that a leaderboard reports, and the state-level router reads per-state means of Y_{im} and per-expert means of c_{im}. The router is built offline and, at test time, chooses an expert before observing any answer, and exactly one expert is invoked per query.

3.2. A unified state-to-expert formulation

This formulation answers the challenge of unknown supervision granularity by writing every router in one form whose granularity is an explicit variable. We consider routers that first assign a query to one of K states and then look the state up in a fixed table. Such a router is the composition

$$r(x) = h(z(x)) \tag{1}$$

of a state assignment

$$z : \mathscr{X} \to \{1, \ldots, K\} \tag{2}$$

and a fixed map

$$h : \{1, \ldots, K\} \to \{1, \ldots, M\}, \tag{3}$$

with output $o(x) = m_{r(x)}(I,t)$. The form costs little generality, since a classifier over experts is the case $K = N$, one state per training sample, with z the identity on the N training samples (the query-level variant below), and it buys two things. It separates the two sources of routing error, assignment error (z places a query in a state whose expert is wrong for it) and estimation error (h stores the wrong expert because the state's competence was estimated from too few or unrepresentative records), so that each can be measured. And it exposes the number of states K, the granularity of the router, as a design variable that need not equal M and may be far smaller.

Because the map must choose one expert per state and must weigh competence against cost, every map considered in this paper is an argmax over experts of a competence estimate minus a cost term,

$$h(k) = \arg\max_{m \in \mathscr{E}_k} \left[\hat{Q}_{km} - \lambda \, \tilde{c}_m \right], \tag{4}$$

where $\hat{Q}_{km}$ estimates the probability that expert m answers a query of state k correctly, $\tilde{c}_m$ is a normalized cost, $\lambda \geq 0$, and $\mathscr{E}_k$ is the set of experts eligible in state k. Equation (4) makes the design space three-dimensional: how z is formed

and hence how large K is, how $\hat{Q}_{km}$ is estimated from the n_k records of a state, and how cost enters (through λ, through $\mathscr{E}_k$, or not at all).

3.3. Cost alongside accuracy

The cost term answers the challenge of accuracy-only expert selection by entering the routing decision of every variant. The objective is the one VL-RouterBench defines for router training [4], expected correctness traded against expected cost with a multiplier λ, and Eq. (4) is its form for a router that consults a state-to-expert map. Where the cost enters depends on the granularity. The query-level variant has one sample per state, so the cost can only enter its labels, which the benchmark's soft labels do by weighting each correct expert by $\exp(-\lambda c_{im})$. The subject-level and state-level variants have states that pool many records, so the cost enters their map as a penalty $\lambda \tilde{c}_m$ subtracted from the competence estimate. Comparing a variant with and without its cost term measures what cost awareness adds, and comparing the variants at the same form measures what granularity adds.

4. Three Bench-CoE variants

4.1. Overview

Constructing a Bench-CoE router has three stages that correspond to the three axes of Eq. (4): form the state assignment z, estimate the competence $\hat{Q}_{km}$ of every expert in every state, and derive the map h with or without a cost term. Online routing is one assignment and one lookup, after which a single expert answers.

The three variants are points of decreasing granularity on the same axis (Fig. 2). Query-Bench-CoE is the finest point. It abstracts the per-query selectors of the routing literature and uses the most specific evidence the records offer, a label for every sample. Subject-Bench-CoE is an intermediate point. It uses a partition that the benchmark already provides, the subjects of a multi-discipline benchmark, and its map is read directly from the published leaderboard. State-Bench-CoE is the coarsest point. It learns a few states from frozen input representations, so that a state exists for any input and holds a large share of the records. All three are built from the same records, each has an accuracy-only and a cost-aware form, and all are evaluated under the same protocol, so the experiments can attribute their differences to granularity and to the cost term rather than to their data.

4.2. Query-Bench-CoE: the query-level variant

The finest choice on the granularity axis makes every benchmark sample its own state, which uses the most specific evidence the records contain. Because several experts may answer the same sample correctly, a sample does not determine one expert, so the label must say how the sample is shared among its correct experts. For sample q_i with ground truth y_i, let $Y_{im} = P(M_m(q_i), y_i) \in \{0,1\}$ score the answer of expert m and c_{im} its cost. The cost-aware routing

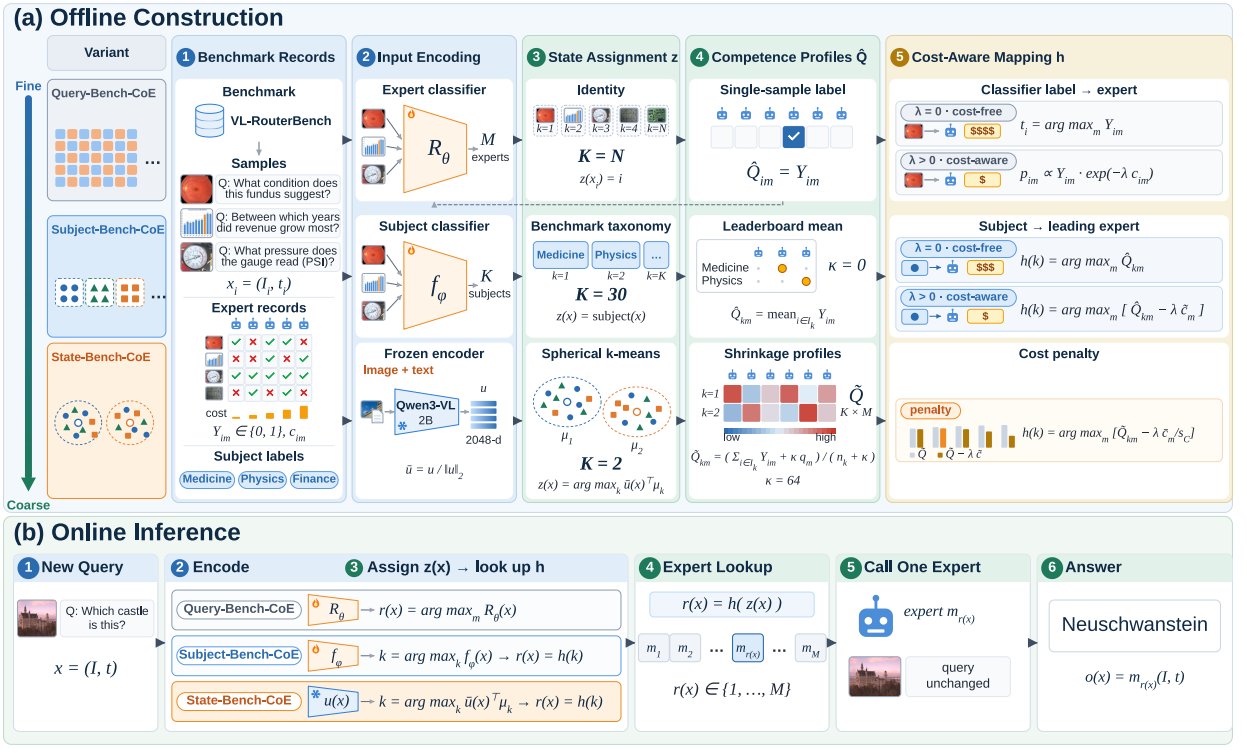


Figure 2: The Bench-CoE framework and its three variants. (a) Offline construction comprises five stages: benchmark records (samples, correctness Y and cost c for each expert, subject labels) $\rightarrow$ input encoding $\rightarrow$ state assignment $z \rightarrow$ competence profile $\hat{Q} \rightarrow$ cost-aware mapping h. The rows correspond to variants of progressively coarser granularity: Query-Bench-CoE, Subject-Bench-CoE and State-Bench-CoE. (b) Online inference consists of a single encoder forward pass, a single state assignment and a single table lookup, followed by a single expert answering the unaltered query.

label is the benchmark's soft label [4], which spreads the sample over its correct experts with weights that decay with cost,

$$p_{im} = \frac{Y_{im} \exp(-\lambda\, c_{im})}{\sum_{m'=1}^{M} Y_{im'} \exp(-\lambda\, c_{im'})}, \qquad Y_{im} = P(M_m(q_i), y_i), \tag{5}$$

where $\lambda \geq 0$ is the cost knob, and as $\lambda \to \infty$ the label concentrates on the cheapest correct expert. The cost-free form replaces Eq. (5) by the hard label $t_i = \arg\max_m Y_{im}$, with ties among correct experts resolved in favour of the expert with the highest overall accuracy on the sample's subject. A classifier R_θ over the M experts is trained on these labels, and inference selects $m = \arg\max R_\theta(q)$ and returns $o = M_m(q)$. The variant abstracts several existing routers [12–14] and coincides with the benchmark's routers trained on hard or soft labels.

Equation (5) needs nothing beyond the records, since every Y_{im} and c_{im} is already published, and its strength is specificity: each decision can reflect the particular sample. Its weakness is the same property. A label is a competence estimate from a single draw, and it binds each decision to one sample rather than to a reusable description of the input,

so the classifier can only interpolate between source samples and has no support where deployment data lie elsewhere. Through Eq. (5) the cost enters the label of every sample, so the trade-off between accuracy and cost is learned query by query.

4.3. Subject-Bench-CoE: the subject-level variant

Multi-discipline benchmarks already partition their samples into subjects and publish per-subject accuracies for many models, and the subject-level variant uses that partition as z and that leaderboard as h. Let the benchmark $B = \{S_1, \ldots, S_K\}$ be partitioned into K subjects. A subject classifier f_φ is trained on the native subject labels, every query in S_k receiving label k. Because the leaderboard already reports each expert's mean score per subject, the map needs only one further quantity, the normalized mean cost $\tilde{c}_m$ of each expert: each subject takes the expert with the highest mean score after a cost penalty,

$$n_k = \arg\max_m \left[\frac{1}{|S_k|} \sum_{i \in S_k} P\left(M_m\left(q_{ki}\right), y_{ki}\right) - \lambda\, \tilde{c}_m \right]. \tag{6}$$

Inference is a subject prediction followed by the lookup: the classifier assigns the query to the subject

$$k = \arg\max f_\varphi(q), \tag{7}$$

and the expert stored for that subject answers,

$$o = M_{n_k}(q). \tag{8}$$

Here $\tilde{c}_m$ is expert m's mean per-query source cost divided by the median positive source cost, and $\lambda \geq 0$ is the cost knob. Equations (6)–(8) are Eq. (4) without shrinkage ($\kappa = 0$ in Eq. (10) below), and $\lambda = 0$ gives the cost-free map, which takes each subject's leaderboard leader. The construction is attractive for two reasons. Its accuracy term is read from a published leaderboard and its cost term needs only each expert's mean cost, so per-query records are not needed. And because the classifier's output layer has K subject neurons rather than M expert neurons, replacing or adding an expert changes only the map. $K \geq M$ is typical, so one expert may serve several subjects.

When a corpus lacks subject labels, as most of VL-RouterBench does, z must be computed by pseudo-labelling. Each training sample receives the subject of its nearest prototype, where the 30 prototypes are the mean Qwen3-VL-2B

image–text features of 735 questions from the official MMMU training split, the only corpus with native labels. The classifier is trained on the pseudo-labels, and the map is built from the source records aggregated by predicted subject. In its cost-free form the map is kept cost-blind and unshrunk, with $\kappa = 0$ and $\lambda = 0$.

4.4. State-Bench-CoE: the state-level variant

State-Bench-CoE keeps the coarseness of subject-level routing but replaces each fixed choice of that variant by a choice made from the data, and three design decisions follow from the bias–variance view of granularity. The states must exist for any input and must not depend on expert outcomes, otherwise they cannot be computed for a new dataset or they overfit to the pool, so we learn them by clustering frozen input representations. The competence of an expert in a state is estimated from the records of that state, and a small state makes this estimate noisy, so we shrink it toward the expert's global accuracy under a prior whose strength has a statistical meaning. The cost must decide among experts whose competences are within estimation error of each other, and it is far less noisy when it acts on pooled estimates than on single labels, so we put it into the map.

4.4.1. Module 1: state construction

The partition must be computable for any input and independent of expert outcomes, and a frozen encoder provides exactly such a representation: it exists without labels and transfers across datasets because it was fitted to none of them. Clustering that representation on source inputs alone therefore yields states that use no correctness or cost record. State-Bench-CoE encodes the image–text pair x with a frozen Qwen3-VL-2B encoder [48], whose pooled 2048-dimensional representation $u(x)$ places both modalities in one space, and L2-normalizes it to $\bar{u}(x) = u(x)/\|u(x)\|_2$. Because normalized features are compared by angle, spherical k-means fits K unit-norm centroids μ_k on a source-fit subset, and a query is assigned to the centroid with the highest cosine similarity,

$$z(x) = \arg\max_{1 \le k \le K} \bar{u}(x)^\top \mu_k, \qquad \|\mu_k\|_2 = 1. \tag{9}$$

No encoder parameter is updated. The number of states K is the granularity of the router. The analysis that follows argues that it should be small under shift, the experiments sweep it, and $K = 2$ is the operating point. Text-only, image-only and Euclidean constructions are compared in the ablation of the state construction.

4.4.2. Module 2: shrinkage-regularized competence profiles

With z fixed, the natural estimate of an expert's competence in a state is its within-state accuracy, and the problem with that estimate is its variance. Let D_p be the profiling subset, here the source-fit subset, $I_k = \{i : x_i \in D_p,\ z(x_i) = k\}$ the samples of state k and $n_k = |I_k|$ its support. The within-state accuracy $\bar{Q}_{km} = n_k^{-1} \sum_{i \in I_k} Y_{im}$ is computed separately

for every expert, so that one sample may support several experts rather than carrying one expert-class label. Its variance is $Q_{km}(1-Q_{km})/n_k$, so in a small state an expert that is best by a few lucky samples is chosen for every query of that state. The global accuracy $q_m = |D_p|^{-1}\sum_i Y_{im}$ is estimated from all source samples and has negligible variance, so we borrow strength from it by adding κ pseudo-samples drawn at the global rate,

$$\tilde{Q}_{km} = \frac{\sum_{i\in I_k} Y_{im} + \kappa q_m}{n_k + \kappa}. \tag{10}$$

The global reference receives more weight when n_k is small, and an empty state inherits q_m. We use $\kappa = 64$, because the profiling set is the source-fit subset, which is smaller than the full source training split and therefore needs more regularization. The shrinkage ablation varies κ.

4.4.3. Module 3: cost-aware mapping

Profiles say which experts are competent in a state, and cost decides among those whose competence is within estimation error of each other. Cost enters the map rather than the labels because a per-query cost weight changes which expert looks best for each sample and is as noisy as the label, whereas a map-level term acts on pooled estimates. Let $\bar{c}_m^{\text{fit}}$ be expert m's mean per-query cost on the source-fit subset. The global mean, not the state-conditional cost, is used because the cost ranking of experts is set by their per-token prices and barely varies across states, so the global mean is a low-variance proxy for the same ranking.

State-Bench-CoE therefore subtracts a cost penalty from the shrunk competence, the Lagrangian form of the trade-off. The penalty keeps every expert eligible and lets a state choose an expensive expert when its competence advantage is large enough,

$$h(k) = \arg\max_m \left[\tilde{Q}_{km} - \lambda\,\frac{\bar{c}_m^{\text{fit}}}{s_C}\right], \tag{11}$$

where s_C is the median positive source cost and $\lambda \geq 0$ is selected on source data. The rule yields a K-entry table that depends on the records only through $\tilde{Q}$ and mean costs, so adding, removing or re-pricing an expert means recomputing h rather than refitting z or any classifier. The cost ablation removes the cost term, and Sec. 7.8 replaces the penalty by a budget that makes experts above the median cost ineligible.

4.4.4. Online inference and complexity

Routing a query costs one encoder forward pass, K inner products, and a table lookup, after which the selected expert processes the original image and text. Router memory is the encoder plus $K \times d$ floats, with d the feature

dimension, and K integers. Relative to end-to-end routers that fine-tune a multimodal backbone [4], the only learned objects are K centroids, and relative to kNN-style routers [14, 40] no training set is retained at inference.

4.5. A post-hoc control layer for attribution

The post-hoc layer is an instrument of the evaluation. It exists to answer one objection: that query- and subject-level routers fail only because their maps ignore cost, so that a cost-aware correction applied afterwards would suffice without changing granularity. The layer wraps any base router. Let a_R be the base router's action and a_A an accuracy candidate, the expert maximizing a local competence estimate $q_A(\cdot)$ with a small cost penalty, where q_A blends a state profile with a nearest-neighbour correction in the source bank (constants in the implementation details). To decide whether a switch from a_R to a_A is worth making, the layer estimates the candidate's gain in competence,

$$g = q_A(a_A) - q_A(a_R), \tag{12}$$

its increase in normalized cost,

$$\Delta c = \tilde{c}(a_A) - \tilde{c}(a_R), \tag{13}$$

and the utility of switching, which weighs the one against the other,

$$u = g - 0.01\,\Delta c, \tag{14}$$

where $\tilde{c}$ is the cost divided by the median positive source cost. A positive utility is necessary for a switch but not sufficient, because the local estimate q_A is only trustworthy for queries that resemble the source data. A confidence score s, the maximum decision score of a ridge classifier trained to recognize a query's source dataset from its frozen features, measures how source-like the query is. Three rules of increasing permissiveness replace a_R by a_A when

R1: $s \geq \tau_{0.80}$ and $u > 0.01$.

R2: $s \geq \tau_{0.35}$ and $u > 0$.

R3: the conditions of R2 and $\Delta c \leq \delta_{0.995}$.

and an optional fallback overrides the resulting action with the expert $h(z(x))$ of a two-state map whenever $s < \gamma$, with γ at the 10% (Guard10) or 25% (Guard25) quantile. All thresholds are quantiles of out-of-fold score distributions on source data, fixed once and not retuned per base router. With the fallback disabled the layer tests whether cost awareness alone transfers, and with it enabled it measures how much of any gain is attributable to the coarse shared-state map.

5. Analysis of granularity, shrinkage and cost

5.1. Shrinkage as a posterior mean

The shrinkage estimate of Eq. (10) is a Bayesian posterior mean, which is why it behaves well on small states. Fix a state k and an expert m, and let Q_{km} be the true probability that m answers a query of state k correctly under the source distribution. Give Q_{km} a Beta$(\kappa q_m, \kappa(1-q_m))$ prior, whose mean is the expert's global accuracy q_m and whose strength is κ pseudo-samples, and let the n_k correctness labels in the state be Bernoulli(Q_{km}). The posterior mean is then

$$\mathbb{E}\left[Q_{km} \mid Y\right] = \frac{\sum_{i \in I_k} Y_{im} + \kappa q_m}{n_k + \kappa} = \tilde{Q}_{km}, \tag{15}$$

which is exactly Eq. (10). Shrinkage is therefore an empirical-Bayes estimator in the sense of Efron and Morris [49]: the global accuracy, estimated from all source samples, serves as the prior mean, and κ trades prior strength against the state's own evidence.

5.2. When shrinkage reduces error

Shrinkage lowers error whenever the state's competence is not too far from the expert's global accuracy, and "too far" has a closed form. Write $\Delta_{km} = q_m - Q_{km}$ for the gap between global and state-specific competence. The unshrunk estimate is unbiased with mean squared error $Q_{km}(1-Q_{km})/n_k$, while the shrunk estimate has

$$\mathrm{MSE}\left(\tilde{Q}_{km}\right) = \frac{n_k Q_{km}\left(1-Q_{km}\right) + \kappa^2 \Delta_{km}^2}{\left(n_k + \kappa\right)^2}. \tag{16}$$

Proposition 1. The shrunk estimate has strictly lower mean squared error than the within-state accuracy if and only if

$$\Delta_{km}^2 < \left(\frac{2}{\kappa} + \frac{1}{n_k}\right) Q_{km}\left(1-Q_{km}\right). \tag{17}$$

Proof. Multiply both sides of Eq. (16) $< Q_{km}(1-Q_{km})/n_k$ by $n_k(n_k+\kappa)^2$, expand $(n_k+\kappa)^2$, cancel $n_k^2 Q(1-Q)$ and divide by $n_k\kappa^2$. □

The condition is easy to satisfy for small states, where $1/n_k$ is large, and for large states it requires only that state-specific competence not depart from global competence by more than about $(2Q(1-Q)/\kappa)^{1/2}$, a departure of about 0.08 for $\kappa = 64$ and $Q \approx 0.7$. Larger κ widens the variance reduction but shrinks the set of states for which the bias term is tolerable, which is why State-Bench-CoE, whose profiling set is the smaller source-fit subset, uses the larger value $\kappa = 64$. The shrinkage ablation reports, for the states actually formed, the fraction of (k,m) pairs for which the condition holds and the effect of κ on the two views.

5.3. Why the number of states should be small under shift

Adding states can raise the variance of the map, because the map is an argmax and an argmax flips when its inputs are noisy. What matters is not the error of each profile entry but whether the estimated ordering between the best expert of a state and its closest competitor is correct. For a state with support $n_k \approx N/K$ and two experts whose true competences differ by a margin η, the standard error of the estimated difference scales as $(K/N)^{1/2}$, so the probability that the estimated argmax differs from the true one grows with K at fixed margin. When the cost term is active, a flipped argmax can substitute an expert several times more expensive.

Under dataset shift [50], map variance can be compounded by thin source coverage, potentially favouring fewer states. A fine partition gives a deployment query a smaller state, so its source support can be thin and the shifted query may resemble none of the records that estimated its profile. A coarse state averages competence over a wider region with more support. Finer states can reduce within-state heterogeneity while increasing estimation and coverage errors, but these empirical terms need not vary monotonically with K. The resulting trade-off is a diagnostic rather than a finite-sample regret bound under shift.

A benchmark taxonomy is the special case in which all three terms are set by annotators rather than by the data. Its states are fixed, so the heterogeneity term is whatever the taxonomy happens to produce on the deployment data. The support of each state is set by the benchmark's subject sizes rather than by K, and no shrinkage or cost term is applied. With 17 experts of widely different prices and deployment data outside the taxonomy, every ordering that flips by chance becomes an expensive choice.

5.4. Map-stability diagnostic

Split the profiling set D_p at random into two halves, build h_A and h_B from each half with the same z and the same rule, and measure (i) the support-weighted fraction of states on which $h_A(k) = h_B(k)$ and (ii) the fraction of test queries that h_A and h_B route to the same expert. Both quantities estimate the variance of the map directly from source data, without target labels, and both should fall as K grows. The diagnostic applies unchanged to a subject taxonomy,

where the subjects play the role of states, and to per-query labels, where it reduces to the agreement of two query-level classifiers trained on disjoint halves.

5.5. The cost term as a Lagrangian

The penalty rule of Eq. (11) is the Lagrangian form of the benchmark's own routing objective, restricted to a fixed partition. VL-RouterBench defines router training as minimizing expected performance risk plus λ times expected cost, $\mathbb{E}[-Y_{i,R(x_i)}] + \lambda\,\mathbb{E}[c_{i,R(x_i)}]$, and derives per-query soft labels from the Lagrangian of this objective [4]. Restricted to a fixed partition the same objective separates over states, and its minimizer is

$$h(k) = \arg\max_{m} \Big[\mathbb{E}\big[Y_{im} \mid z(x_i) = k\big] - \lambda\,\mathbb{E}\big[c_{im} \mid z(x_i) = k\big]\Big], \tag{18}$$

which is Eq. (11) with the state-conditional cost replaced by the expert's global mean cost. Per-state cost estimates inherit the variance of output lengths, whereas the cost ranking of experts is dominated by their per-token prices and is nearly invariant across states, so the global mean is a low-variance proxy for the same ranking. The cost ablation substitutes the state-conditional cost back in to measure the difference. A budget that makes expensive experts ineligible is the constrained form of the same problem, and Sec. 7.8 compares it with the penalty.

6. Experimental setup

6.1. Integrated benchmark and its records

All routers are built and evaluated on VL-RouterBench [4], an integrated benchmark that gathers 14 public datasets in three task groups, general, STEM and charts with OCR, and scores 15 open-source models, such as Qwen2.5-VL, and 2 API models, such as GPT-4o, on all of their 30,540 single-image samples. Its release contains 519,180 records, one per sample and expert, each with a rule-based binary correctness label and the token counts of the expert's answer. Cost follows from these counts and the published per-million-token prices, which span multiple orders of magnitude across the pool [4]. The records therefore contain every quantity that any Bench-CoE variant needs, and no router required running an expert or labelling a query. VL-RouterBench is chosen because it is the only benchmark that combines a large, cost-heterogeneous pool of vision–language experts with a protocol that withholds whole datasets from router construction. On it the oracle that selects the cheapest correct expert per sample reaches 95.6% accuracy at 0.37 USD per 10K samples over all 14 datasets [4]. The benchmark's 7:1:2 train, validation and test split is used throughout.

6.2. Evaluation protocols

Two views are reported for every method. They follow the benchmark's out-of-distribution (OOD) setting, which withholds one dataset per task group, MMMU, MathVision and ChartQA, from router construction. 11-ID is the in-distribution (ID) view on the test splits of the eleven seen datasets (5,341 queries) and 3-OOD the OOD view on the test splits of the three withheld ones (767 queries). Both share one frozen router built from the training splits of the eleven seen datasets, and 3-OOD is the primary evidence for transfer. Every method is scored by its accuracy (A, in %), its mean expert cost (C, in USD per 10K queries, excluding the router's own overhead) and the Rank Score (R), the weighted harmonic mean of accuracy and log-normalized cost with $\beta = 0.1$ defined by VL-RouterBench [4].

6.3. Variants and reference routers

6.3.1. The three Bench-CoE variants

All three variants are built from the same source records. Query-Bench-CoE and Subject-Bench-CoE are trained with the TinyLLaVA-Phi-2-SigLIP-3.1B [51] classifier backbone on the 11-dataset training split for 11-ID/3-OOD. The subject-level router uses the prototype pseudo-labels described earlier. The cost-free form, with hard labels at the query level and $\kappa = 0$ and $\lambda = 0$ in Eq. (6) at the subject level, holds the accuracy ceiling, and its transfer is what the paper diagnoses. The cost-aware form has one cost knob each, treated exactly like the reference routers': the soft labels of Eq. (5) over the benchmark's λ grid at the query level and the penalized map of Eq. (6) at the subject level, each at the operating point selected on source data. All comparisons with the reference routers use the TinyLLaVA backbone for both the query level and the subject level. Because the subject-level prototypes are built from MMMU training inputs and MMMU's test split is part of 3-OOD, its 3-OOD figures carry an input-side advantage (Sec. 8.6). State-Bench-CoE uses the multimodal spherical construction, alternative constructions are compared in Sec. 7.8, and the post-hoc control layer is applied to the cost-free query- and subject-level routers as base routers, to attribute whatever transfer it produces.

6.3.2. Reference routers published with the benchmark

The ten routers released with VL-RouterBench serve as external reference points, since they use the same benchmark records. Feature-level routers are KNN [14], PRkNN, OVR, K-means, Linear and MLP; end-to-end routers are CosineCls, RouterDC [15], ZOOTER [13] and VLC [12]. ZOOTER appears only in the paired bootstrap comparison of Sec. 7.7. For each reference method and view, Table 1 uses the aggregate printed in VL-RouterBench wherever it exists, which covers the 11-ID/3-OOD Oracle, RouterDC, VLC and MLP values [4, Table A2]. VL-RouterBench selected its reported operating points on test Rank Score. For the remaining 11-ID/3-OOD methods, which VL-RouterBench does not report, we use the released benchmark records and archived per-query actions or retrained models, as marked in Table 1. Published and locally measured operating points therefore have different

selection and run conventions, and Table 1 marks the provenance of every value. Oracle is a test-outcome upper bound. Strongest calls the highest-source-accuracy expert and Cheapest calls the least expensive expert.

6.4. Implementation details

State-Bench-CoE uses L2-normalized Qwen3-VL-2B pooled features (2048-d), spherical k-means on the source-fit subset, $\kappa = 64$, and the global cost penalty with s_C the median positive source cost and λ selected by group-isolated three-fold cross-validation on source data, maximizing source Rank Score, which gives $\lambda = 0.03$. The text-only construction of the ablation uses all-MiniLM-L6-v2 [52, 53] (384-d), spherical k-means, $\kappa = 20$ and the median-cost eligibility filter at quantile 0.5, selected by source cross-validation from $\{0.25, 0.35, 0.5, 0.65, 0.75, 1\}$. K is swept over $\{1, 2, 4, 6, 8, 10, 12\}$ in the state-count experiment, each K over five seeds with λ re-selected on source data, which gives $\lambda = 0.03$ up to $K = 10$ and $\lambda = 0.1$ at $K = 12$, and $K = 2$ is the reported operating point. State-Bench-CoE results are five-seed means, and its paired bootstrap tests use the seed-0 routes. The post-hoc control layer uses an accuracy candidate that blends a state profile with a nearest-neighbour correction over the source bank in TinyLLaVA feature space (11-source: blend 0.8/0.2, 32 same-state neighbours, softmax temperature 0.08, cost weight 0.025). Its confidence score is the maximum decision score of a ridge classifier over source-dataset identity on the same features. Its thresholds are quantiles of three-fold, group-isolated out-of-fold scores on the 16,269 source-fit samples, $\tau_{0.80} = 0.932$, $\tau_{0.35} = 0.551$, $\delta_{0.995} = 3.221$ (normalized cost units), $\gamma_{10} = 0.146$ and $\gamma_{25} = 0.435$, and its fallback expert is that of the frozen two-state 11-source map of that text-only construction. The thresholds were selected once for State-Bench-CoE's own candidate rules and applied to the base routers without retuning.

7. Results

7.1. Overall comparison of the three variants

The ranking of routers in distribution does not predict their ranking on OOD datasets (Table 1). On the eleven seen datasets State-Bench-CoE leads the displayed deployable routers at a Rank Score of 74.09 and 0.65 USD per 10K queries. The cost-aware Query-Bench-CoE and Subject-Bench-CoE with TinyLLaVA follow at 73.83, but the query router uses 1.63 USD and the subject router 0.66. On the three OOD datasets State-Bench-CoE leads at 73.09, followed by CosineCls at 71.18 and Subject-Bench-CoE at 70.58, while Query-Bench-CoE falls to eighth place at 63.32. The 11-ID to 3-OOD losses are 1.00, 3.25 and 10.52 Rank points for the state, subject and query levels.

The published OOD comparison covers RouterDC, VLC and MLP. Their Rank Scores change from 73.26 to 68.88, 72.07 to 69.44, and 70.66 to 55.79 between 11-ID and 3-OOD, respectively. Their losses of 4.38, 2.63 and 14.87 points exceed State-Bench-CoE's 1.00-point loss. The best published 3-OOD reference score is VLC's 69.44,

Table 1
Results in the ID and OOD settings. A is accuracy (%), C expert cost (USD per 10K queries), and R Rank Score. † transcribes [4, Table A2], at that paper's reported operating point; ‡ is locally evaluated from released records because VL-RouterBench has no corresponding value. Bench-CoE rows use source-selected operating points. Published costs retain VL-RouterBench's two-decimal precision; local costs use three decimals. ΔR is 11-ID minus 3-OOD, computed from the unrounded Rank Scores, so it can differ by 0.01 from the difference of the displayed values. Bold marks the best deployable router per metric and view.

Method	11-ID A	11-ID C	11-ID R	3-OOD A	3-OOD C	3-OOD R	ΔR
Oracle (upper bound) †	94.85	0.39	92.87	96.78	0.47	93.94	−1.07
Strongest (Qwen2.5-VL-72B) ‡	78.28	2.689	69.18	78.62	2.870	68.75	0.43
Cheapest (DeepSeek-VL2-Tiny) ‡	62.46	0.144	64.67	65.19	0.147	67.31	−2.64
K-means [4] ‡	72.74	1.885	67.77	39.50	1.727	39.74	28.03
KNN [14] ‡	67.01	**0.384**	67.83	65.45	0.490	65.96	1.87
Linear [4] ‡	72.81	1.473	69.15	62.58	2.162	58.90	10.25
CosineCls [15] ‡	71.17	0.711	70.33	70.80	0.419	71.18	−0.85
OVR [4] ‡	76.95	1.945	70.85	54.11	3.545	49.12	21.73
RouterDC [15] †	73.62	0.53	73.26	68.81	0.54	68.88	4.38
PRkNN [4] ‡	71.99	0.427	72.24	67.54	**0.403**	68.24	4.00
VLC [12] †	72.09	0.49	72.07	69.36	0.52	69.44	2.63
MLP [4] †	71.61	0.73	70.66	55.53	0.86	55.79	14.87
Query-Bench-CoE, TinyLLaVA, cost-aware (soft labels)	**79.33**	1.631	73.83	68.45	2.246	63.32	10.52
Subject-Bench-CoE, TinyLLaVA, cost-aware (λ-penalized map)	74.89	0.661	73.83	71.32	0.681	70.58	3.25
State-Bench-CoE ($K = 2$)	75.12	0.647	**74.09**	**73.92**	0.633	**73.09**	1.00

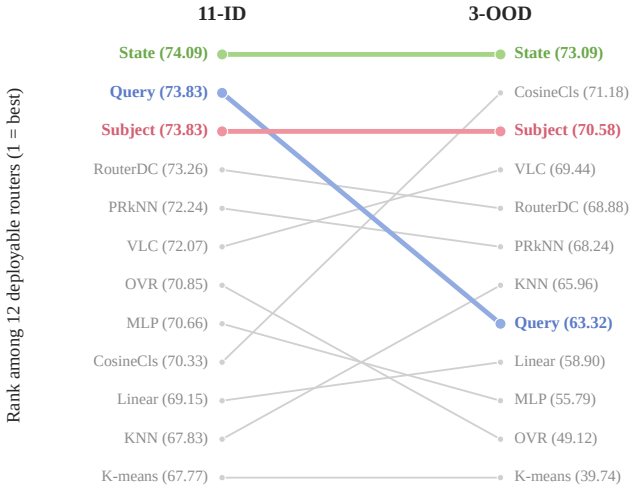


Figure 3: The three variants and the reference routers using Table 1 values. Rank of the twelve deployable routers by Rank Score on 11-ID and 3-OOD, with the Rank Score beside each name. Published reference values are used where available, and local replays fill unreported OOD views.

3.65 points below State-Bench-CoE. The other reference rows in Table 1 use local replays because VL-RouterBench does not report those views. Paired tests of those frozen local actions are reported in Sec. 7.7.

Fig. 3 draws the rank of every deployable router on 11-ID and 3-OOD using the values of Table 1. Published reference values are used where available, and clearly marked local replays fill the unreported OOD views. State-Bench-CoE ranks first on both views, while the TinyLLaVA cost-aware query router loses 10.52 Rank points.

Table 2
The cost term across variants: each variant with and without its cost term, with TinyLLaVA at the query and subject levels, and the cost estimate of State-Bench-CoE ablated. Cells are A / C / R; bold marks the highest Rank Score per view.

Method	Cost term	Cost handling	11-ID A / C / R	3-OOD A / C / R
Query-Bench-CoE, TinyLLaVA	✗	none (hard labels)	78.84 / 2.768 / 69.29	78.88 / 2.974 / 68.54
Query-Bench-CoE, TinyLLaVA	✓	soft labels, Eq. (5), source-selected λ	79.33 / 1.631 / 73.83	68.45 / 2.246 / 63.32
Subject-Bench-CoE, TinyLLaVA	✗	none ($\kappa = 0$, $\lambda = 0$)	81.22 / 3.076 / 69.73	59.32 / 4.273 / 51.00
Subject-Bench-CoE, TinyLLaVA	✓	λ-penalized map, Eq. (6), source-selected λ	74.89 / 0.661 / 73.83	71.32 / 0.681 / 70.58
State-Bench-CoE ($K = 2$)	✓	global penalty, Eq. (11) (full)	**75.12 / 0.647 / 74.09**	**73.92 / 0.633 / 73.09**
State-Bench-CoE ($K = 2$)	✗	none ($\lambda = 0$)	79.15 / 1.978 / 72.42	75.88 / 1.973 / 69.93
State-Bench-CoE ($K = 2$)	✓	state-conditional cost in the penalty	**75.12 / 0.647 / 74.09**	**73.92 / 0.633 / 73.09**

7.2. Query-Bench-CoE

Per-query supervision fits the source datasets, and without a cost term it reproduces the strongest expert (Table 2). With the TinyLLaVA backbone the cost-free query-level router reaches 78.84% accuracy at 2.77 USD and a Rank Score of 69.29 on 11-ID, within 0.6 points of the strongest expert at 78.28%, 2.69 USD and 69.18, and on 3-OOD it reaches 78.88% at 2.97 USD and 68.54 against 78.62%, 2.87 USD and 68.75 for that expert. The reason is the label. A classifier trained to name the best expert of each sample learns to name the expert that is most often correct, so it gains no accuracy from routing, saves no cost and transfers exactly as well as that expert does.

The cost term improves the TinyLLaVA query-level router on seen datasets but does not transfer under shift (Table 2). With source-selected soft labels it reaches 79.33% accuracy, 1.631 USD and a Rank Score of 73.83 on 11-ID, compared with 78.84%, 2.768 USD and 69.29 without the cost term. On 3-OOD it falls to 68.45% and 63.32, against 78.88% and 68.54 in the cost-free form. The 10.52-point Rank Score drop from 11-ID to 3-OOD comes from a per-query trade-off learned on source records. On ChartQA, 62.5% of queries are routed to an API or 70B-class expert, and the cost-aware router reaches 71.76% accuracy, 1.440 USD and a Rank Score of 68.39. Choosing λ on the OOD test set would lift the pooled 3-OOD score to 68.61 at $\lambda = 10000$, but that test-selected operating point is only a diagnostic.

The conclusion for Query-Bench-CoE is that per-query records can improve the seen-dataset accuracy–cost trade-off, but the source-selected cost-aware TinyLLaVA router remains 0.26 Rank points below State-Bench-CoE on 11-ID and loses 10.52 points under shift. Its accuracy-only form adds little over the strongest expert.

7.3. Subject-Bench-CoE

Subject-level supervision holds the accuracy ceiling at a high price, and without a cost term it collapses under shift (Table 2). In distribution the cost-free subject-level router with the TinyLLaVA backbone is the most accurate router on the benchmark, 81.22% on 11-ID, 2.9 points above the strongest expert. It pays with the highest cost of any router, 3.08 USD on 11-ID, four to six times the published costs of VLC, MLP and RouterDC, and its Rank Score of 69.73 on 11-ID is 0.93 points below the published MLP score. On the three OOD datasets its accuracy falls to 59.32% at

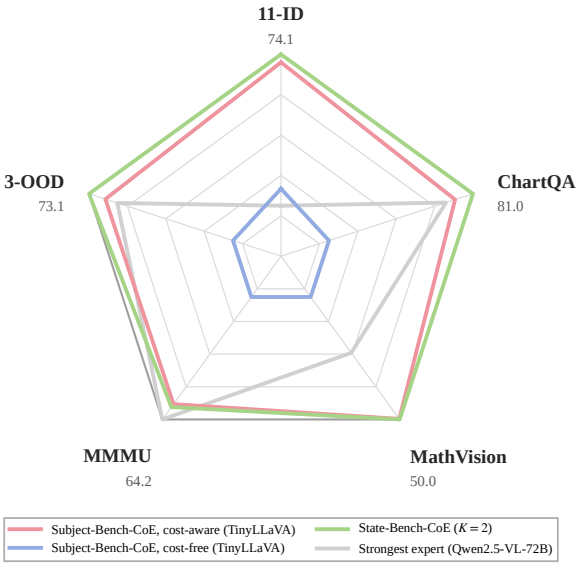


Figure 4: Rank Score of Subject-Bench-CoE, State-Bench-CoE and the strongest expert on the two views and the three OOD datasets. Each axis has its own scale, from the lowest value of the four plotted routers near the centre to the highest on the outer ring, and that highest value is printed under the axis name. Colours follow Fig. 3: the light pink polygon is the cost-aware subject-level form and the light blue polygon the cost-free form, both with the TinyLLaVA backbone, the light green polygon is State-Bench-CoE ($K = 2$) and the light grey polygon is the strongest expert.

4.27 USD and a Rank Score of 51.00, far below the cheapest expert at 67.31. This happens even though its prototypes were built from MMMU training inputs and MMMU is one of the OOD datasets.

Fig. 4 presents the Rank Score of the subject-level variant, beside State-Bench-CoE and the strongest expert, across the two views and the three OOD datasets, in radar form. The cost-free router collapses toward the centre on MMMU and MathVision, down to 0 on MathVision, even though it is the most accurate router on the seen datasets and on MMMU, where it reaches 74.8%, because that accuracy is bought with API and 70B-class experts. The cost-aware form keeps a nearly regular polygon, with Rank Scores between 49.8 and 77.7 on every axis. State-Bench-CoE is the outermost polygon on every axis except MMMU, where the strongest expert leads, and stays 2.5 Rank points above the cost-aware form on 3-OOD (Table 1).

The cost term repairs most of this collapse, and the remainder belongs to the partition. The λ-penalized map recovers 19.6 of the TinyLLaVA router's OOD Rank points, from 51.00 to 70.58, at a cost of 0.68 USD, and leaves a gap of 2.5 points to State-Bench-CoE. The mechanism is the one the analysis describes. Thirty cost-blind, unshrunk per-subject maps, each estimated from a few dozen to a few hundred records, select an expensive expert whenever an ordering flips by chance, and the cost term removes the expensive experts from most of those flips.

The conclusion for Subject-Bench-CoE is that a published taxonomy gives free and accurate supervision, but its accuracy-only map is too costly to deploy across a cost-heterogeneous pool and too noisy to transfer. With a cost term

Table 3
Effect of the number of states K on State-Bench-CoE. Cells are A / C / R on both views. $K = 1$ is a single fixed expert, and values are five-seed means with a source-selected λ for each K. Bold marks the highest Rank Score on each view.

Implementation	K	11-ID A / C / R	3-OOD A / C / R
State-Bench-CoE	1	74.57 / 0.76 / 73.14	67.54 / 0.71 / 67.10
State-Bench-CoE	2	75.12 / 0.65 / 74.09	73.92 / 0.63 / 73.09
State-Bench-CoE	4	75.02 / 0.63 / 74.07	**73.92 / 0.62 / 73.13**
State-Bench-CoE	6	76.25 / 0.79 / 74.49	74.58 / 0.80 / 72.98
State-Bench-CoE	8	76.28 / 0.79 / 74.49	73.14 / 0.83 / 71.63
State-Bench-CoE	10	**77.38 / 0.97 / 74.72**	74.99 / 1.85 / 69.66
State-Bench-CoE	12	74.50 / 0.57 / 73.89	72.88 / 0.61 / 72.24

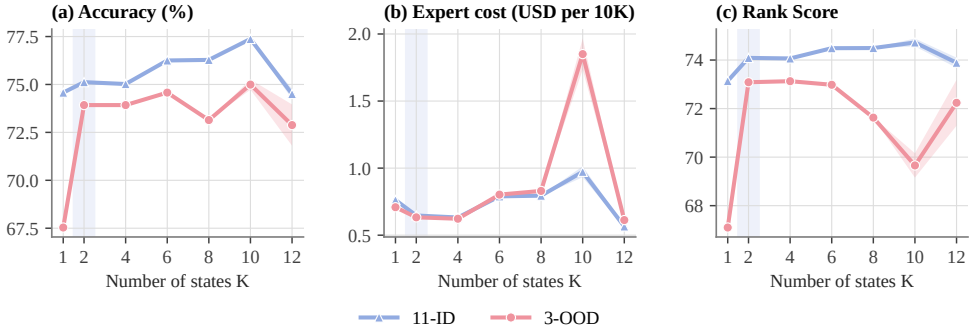


Figure 5: Effect of the number of states K on (a) accuracy, (b) expert cost and (c) Rank Score of State-Bench-CoE, as five-seed means with a band of one standard deviation. The shaded column marks the operating point $K = 2$.

in its map it comes within 0.3 Rank points of the state-level variant in distribution and stays 2.5 points below it on the OOD datasets.

7.4. State-Bench-CoE

A few learned states transfer best, and moving from one shared state to two is the decisive step (Table 3, Fig. 5). With $K = 1$ the router is a fixed expert chosen by the cost rule, Kimi-VL-A3B-Thinking, which already beats most reference routers in distribution but reaches only 67.10 on 3-OOD. Two states raise OOD accuracy by 6.38 points and OOD Rank Score by 5.99, while lowering cost. Beyond two states the two views part. $K = 4$ matches $K = 2$ within 0.04 Rank points on both views at slightly lower cost. From $K = 6$ to $K = 10$ finer states raise in-distribution accuracy to 77.38% and the 11-ID Rank Score to 74.72, but out of distribution the cost rises to 1.85 USD at $K = 10$ and the Rank Score falls to 69.66, so finer states buy accuracy by substituting expensive experts. At $K = 12$ source cross-validation selects the larger $\lambda = 0.1$, which restores a low cost and gives 72.24 on 3-OOD. The best OOD trade-off therefore lies at $K = 2$ to 4, and $K = 2$ is the operating point because the map-stability diagnostic, measured on the Euclidean partitions of Sec. 7.5, favours the smallest K.

At two states the map is insensitive to the shrinkage strength, and the condition of Proposition 1 holds for most state–expert pairs at the default κ (Table 4). For every κ from 5 to 512 the shrunk profiles choose the same experts in

Table 4
Shrinkage strength κ at $K = 2$: five-seed mean support-weighted fraction of source state–expert pairs satisfying Proposition 1. The $\kappa = 0$ fraction is undefined, and map invariance is reported in the text.

κ	State-Bench-CoE, source-fit fraction satisfying Prop. 1
0 (no shrinkage)	—
5	1.000
20	0.909
64	0.775 (default)
128	0.641
512	0.264

all five seeds, so the results are those of the operating point. At $\kappa = 64$ the condition under which shrinkage lowers mean squared error holds, weighted by support, for 77.5% of source state–expert pairs, against 90.9% at $\kappa = 20$, and the fraction falls to 26.4% at $\kappa = 512$. Shrinkage is therefore insurance for small states rather than a driver of the two-state results, and it is what keeps the thirty-state maps of Sec. 7.5 usable.

The cost term adds transfer to a partition that already transfers (Table 2). Without its cost term State-Bench-CoE gains 4.0 points of accuracy on 11-ID and 2.0 on 3-OOD and roughly triples its cost, to 1.98 and 1.97 USD, and it is still the best cost-free router out of distribution, 69.93 against 68.54 for the query level and 51.00 for the subject level. The coarse partition therefore transfers on its own, and the cost term adds 1.67 Rank points on 11-ID and 3.16 on 3-OOD. Replacing the global mean cost by the state-conditional mean changes nothing.

The conclusion for State-Bench-CoE is that a few states learned from frozen features, with pooled competence profiles and cost in the map, give the most transferable router of the three variants. Two states are enough, four do no better, and shrinkage protects the small states of finer partitions.

7.5. Disentangling granularity and the cost rule

At fixed granularity the cost-aware rule contributes most of the transfer, and a learned partition contributes the rest (Table 5, Fig. 6). The matched-granularity comparison holds the source records fixed and varies the partition, the 30 MMMU subjects by nearest prototype, Euclidean k-means with $K = 30$ and with $K = 2$, all in the same Qwen3-VL feature space. Each partition is combined with two mapping rules, the plain argmax of the subject-level variant and the shrunk, cost-penalized rule of State-Bench-CoE with λ re-selected on source data for each partition. Under the plain rule granularity is decisive. The OOD Rank Score rises from 56.39 for the taxonomy to 60.17 for thirty learned states and 69.65 for two, because every ordering that flips by chance in a small, cost-blind state selects an expensive expert. Under the cost-penalized rule the same partitions reach 71.34, 73.04 and 72.72. The rule thus lifts the taxonomy by 15.0 points, and the learned partition adds 1.7 points to that. Thirty learned states perform like two on this seed, and the stability diagnostic below shows why two remain the safe choice. With spherical instead of Euclidean clustering the two-state router reaches 73.09 on 3-OOD (Sec. 7.8).

Table 5
Matched-granularity comparison: the partition z varies while the mapping rule is held fixed, all in Qwen3-VL-2B feature space. Cells are A / C / R. Bold marks the highest Rank Score per view. Learned partitions use Euclidean mini-batch k-means.

Partition z	K	Shrinkage ($\kappa > 0$)	Cost term ($\lambda > 0$)	11-ID A / C / R	3-OOD A / C / R
Taxonomy (MMMU subjects)	30	✗	✗	81.03 / 2.985 / 69.96	70.14 / 4.691 / 56.39
Taxonomy (MMMU subjects)	30	✓	✓	74.56 / 0.628 / 73.67	71.97 / 0.634 / 71.34
Learned (Euclidean k-means)	30	✗	✗	81.28 / 2.703 / 71.24	73.14 / 4.188 / 60.17
Learned (Euclidean k-means)	30	✓	✓	74.54 / 0.569 / 73.90	**73.79 / 0.615 / 73.04**
Learned (Euclidean k-means)	2	✗	✗	78.88 / 1.943 / 72.34	75.23 / 1.907 / 69.65
Learned (Euclidean k-means)	2	✓	✓	**75.15 / 0.651 / 74.10**	73.53 / 0.638 / 72.72

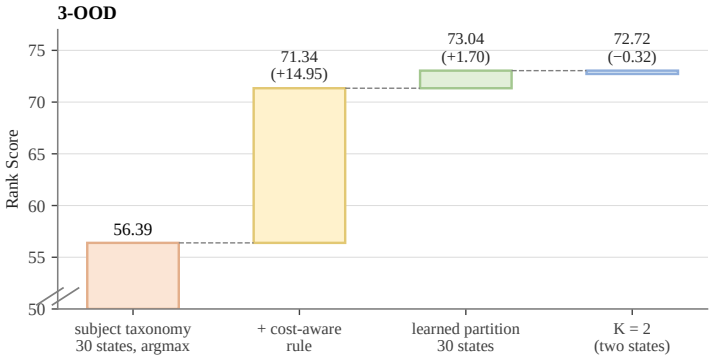


Figure 6: Contribution of the cost-aware rule and the partition to the Rank Score on 3-OOD, read as a ladder from the cost-free subject taxonomy to two Euclidean k-means states under the State-Bench-CoE rule.

The agreement of a map with itself across disjoint halves of the source data falls with the number of states, as the analysis predicts, and only the two-state map is fully reproducible. With the source records split at random into two halves and a map built from each, over ten splits with the same partition and rule, the two-state map agrees with itself on 100% of its support and routes every 11-ID and 3-OOD query identically. Agreement falls to 98% at $K = 4$, 77% at $K = 12$ and 52% at $K = 30$, and the routing agreement on OOD queries falls faster still, to 48% and 29%, because an OOD query is more likely to land in a state whose expert the two halves disagree about. The subject taxonomy, with as many states as the thirty-state partition, is more stable at 74% because its states are larger and more even, and less stable than any learned partition with $K \leq 12$. Per-query classifiers trained on disjoint halves agree on 99% of queries, which reflects their common convergence to the strongest expert rather than a stable map. The diagnostic uses no target labels and reproduces the ordering of the state-count sweep, so it can set K before deployment.

7.6. Attribution with the post-hoc control layer

When a fine router is combined with the coarse map, the coarse map is the part that transfers (Table 6, Fig. 7). The post-hoc control layer, with thresholds fixed from the state-level router's own source statistics, is applied to

Table 6
Post-hoc cost control with rule R3 and the Guard25 fallback, applied to the cost-free fine routers with the TinyLLaVA backbone on the 11-source model. Cells are A / C / R, and the last two rows apply the layer to its fallback, the two-state text-only map of Sec. 7.8.

Base router	Control	11-ID A / C / R	3-OOD A / C / R
Query-Bench-CoE, TinyLLaVA	none	78.84 / 2.768 / 69.29	78.88 / 2.974 / 68.54
Query-Bench-CoE, TinyLLaVA	R3 + Guard25 fallback	79.50 / 1.712 / 73.67	74.84 / 1.110 / 72.06
Subject-Bench-CoE, TinyLLaVA	none	81.22 / 3.076 / 69.73	59.32 / 4.273 / 51.00
Subject-Bench-CoE, TinyLLaVA	R3 + Guard25 fallback	79.84 / 1.931 / 73.12	69.10 / 1.121 / 67.15
Two-state text-only map, fallback	none (reference)	75.55 / 0.656 / 74.43	74.19 / 0.653 / 73.24
Two-state text-only map, fallback	R3 + Guard25 fallback	76.54 / 0.721 / 75.02	74.19 / 0.653 / 73.24

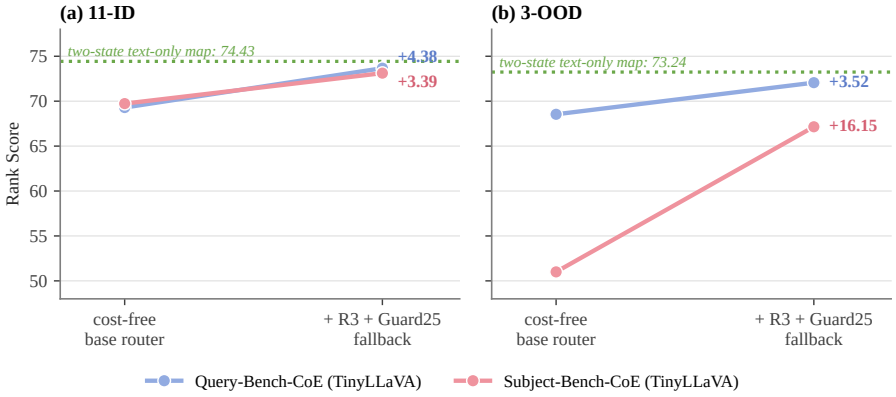


Figure 7: The post-hoc control layer as a paired change for the cost-free query- and subject-level routers with the TinyLLaVA backbone on (a) 11-ID and (b) 3-OOD. The dotted line is the two-state text-only fallback map without the layer.

the cost-free query- and subject-level routers with the TinyLLaVA backbone on the 11-source model, with the most permissive rule R3 and the Guard25 fallback.

The fallback is what transfers. On the OOD datasets the Guard25 gate fires on 81.4% of queries for every base router, and the fallback changes the action on 79–81% of them, so four fifths of the hybrids' OOD decisions are those of the two-state text-only map that serves as the fallback. Accordingly, the query-level hybrid lands at 72.06 on 3-OOD, 1.2 Rank points below that map at 73.24, and the subject-level hybrid at 67.15. Cost awareness at the original granularity does not substitute for this. The cost-aware forms of the same variants in Table 1, which put the cost into the labels or into the subject map, reach 63.32 and 70.58 on 3-OOD.

The hybrids occupy a different operating point rather than a dominated one. In distribution they keep 79.50% and 79.84% accuracy, four points above the two-state text-only map, at 1.71 and 1.93 USD, and the same layer lifts that map itself to 75.02 in distribution at 0.72 USD without changing any of its OOD decisions.

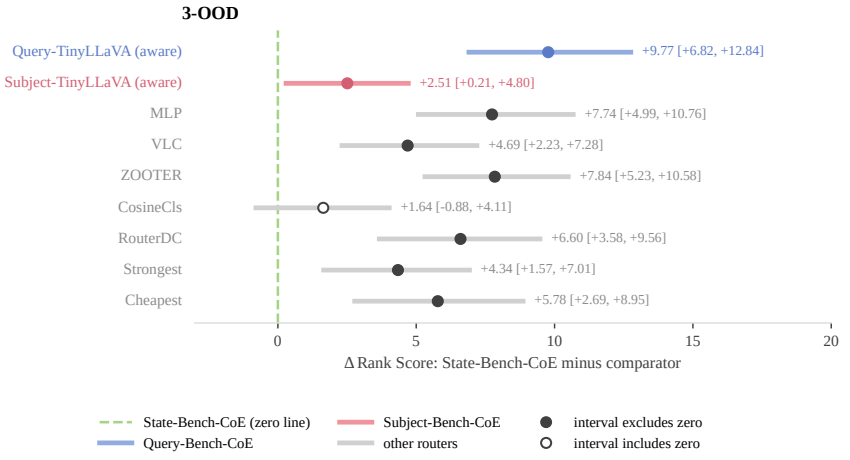


Figure 8: Paired bootstrap differences in Rank Score on 3-OOD between State-Bench-CoE and each comparator, using local frozen seed-0 per-query actions, with 95% intervals. The Query-Bench-CoE row is blue, the Subject-Bench-CoE row pink and the other comparators grey, the dashed green zero line stands for State-Bench-CoE, and filled points mark intervals that exclude zero. These intervals do not test the published aggregates in Table 1.

7.7. Variability and significance

Using the locally frozen seed-0 per-query actions, the OOD advantage of State-Bench-CoE over the other two variants and over every reference router except CosineCls is significant, while in distribution it trails only the local MLP replay, by 0.98 Rank points (Fig. 8). On 3-OOD, 2,000 paired bootstrap resamples put State-Bench-CoE 9.77 points above cost-aware Query-Bench-CoE with TinyLLaVA, with a 95% interval of 6.82–12.84 and $p < 0.001$, and 2.51 points above cost-aware Subject-Bench-CoE, with $p = 0.032$, as well as 4.3–7.8 points above MLP, VLC, RouterDC, ZOOTER and the strongest expert. Its 1.64-point margin over CosineCls does not exclude zero, $p = 0.212$. On 11-ID its leads of 0.25 over the TinyLLaVA query router and 0.26 over the subject router do not exclude zero, $p = 0.650$ and $p = 0.319$, and it trails the local MLP replay by 0.98, $p = 0.028$. Across five seeds the Rank Score of State-Bench-CoE varies by a standard deviation of 0.007 on 11-ID and not at all on 3-OOD, against 0.7 for cost-free query classifiers and 1.7 for cost-free subject classifiers out of distribution in their five-seed runs.

7.8. Ablation of the state construction and the cost rule

Cosine assignment on the multimodal features transfers slightly better than Euclidean assignment, and the transferable signal lies in the question text rather than in the image (Table 7, Fig. 9). On the same Qwen3-VL-2B features, spherical k-means with cosine assignment and mini-batch k-means with Euclidean assignment reach the same 11-ID accuracy of 75.12%, but the spherical states lower the 3-OOD cost from 0.637 to 0.633 USD and raise the 3-OOD Rank Score from 72.84 to 73.09, so the loss under shift falls from 1.24 to 1.00 points.

Table 7
Ablation of the state construction of State-Bench-CoE at $K = 2$. Cells are A / C / R at the measured precision, and $\Delta R =$ 11-ID − 3-OOD. The first row is the main construction. Bold marks the highest Rank Score per view. The MiniLM row uses the eligibility filter rather than the penalty.

State construction	Input	11-ID A / C / R	3-OOD A / C / R	ΔR = 11-ID − 3-OOD
Qwen3-VL-2B, spherical k-means, cosine assignment, main	image + text	75.121 / 0.647 / 74.089	73.924 / 0.633 / 73.086	1.003
Qwen3-VL-2B, mini-batch k-means, Euclidean assignment	image + text	75.121 / 0.650 / 74.074	73.664 / 0.637 / 72.837	1.237
MiniLM-L6-v2	text	75.533 / 0.656 / 74.414	74.185 / 0.653 / 73.235	1.179
BGE-M3	text	75.473 / 0.648 / 74.395	**75.098 / 0.635 / 74.120**	0.275
SigLIP	image	74.649 / 0.748 / 73.257	68.188 / 0.702 / 67.706	5.551
BGE-M3 + SigLIP	image + text	**75.585 / 0.647 / 74.497**	74.576 / 0.634 / 73.660	0.837

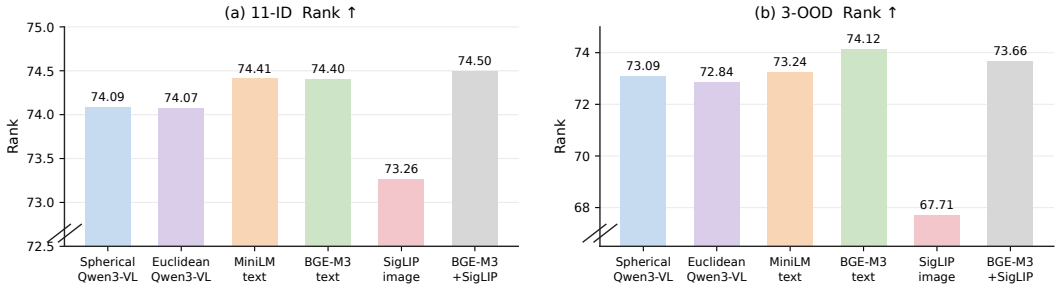


Figure 9: Rank Score of State-Bench-CoE under the six state constructions of Table 7 on (a) 11-ID and (b) 3-OOD. The y-axes are truncated.

Text alone carries most of the transfer, and the image alone does not. The MiniLM-L6-v2 text construction reaches 74.41 on 11-ID and 73.24 on 3-OOD, and BGE-M3 text features reach 74.40 and 74.12, the smallest loss under shift of any row at 0.28 points. SigLIP image features reach 73.26 on 11-ID but fall to 67.71 on 3-OOD, close to the single fixed expert at 67.10, so states formed from images alone do not separate regions of expert competence that hold on new datasets. Combining BGE-M3 with SigLIP gives the highest 11-ID Rank Score, 74.50, and 73.66 on 3-OOD.

Two alternatives exceed the main construction on 3-OOD, BGE-M3 text-only by 1.03 points and BGE-M3 with SigLIP by 0.57. The conclusions of the main text therefore do not rest on the strongest available encoder, and a stronger text encoder is the most direct way to improve State-Bench-CoE further.

The penalty is kept in the main text because it is the more robust of the two cost rules, not because it scores higher at the operating point (Table 8). With the global mean cost the eligibility filter and the penalty select the same experts at two states, so swapping them leaves all three constructions unchanged, the spherical main construction included at 74.09 and 73.09. The rules part when the global mean cost is replaced by the state-conditional mean. The penalty still leaves the spherical and the Euclidean multimodal routers unchanged, whereas the filter collapses the text-only router to a single expert, Qianfan-VL-8B, and loses 3.83 Rank points on 11-ID and 3.95 on 3-OOD, because a hard threshold is sensitive to the within-state cost noise that the global mean averages out.

Table 8
Ablation of the cost rule of State-Bench-CoE at $K = 2$. Cells are A / C / R. Rows of the Euclidean and MiniLM constructions are seed-0 measurements, so they differ slightly from the five-seed means of Table 7.

State construction	Cost rule and cost estimate	11-ID A / C / R	3-OOD A / C / R
Qwen3-VL-2B, spherical, main	penalty, global mean cost	75.12 / 0.647 / 74.09	73.92 / 0.633 / 73.09
Qwen3-VL-2B, spherical, main	eligibility filter, global mean cost	75.12 / 0.647 / 74.09	73.92 / 0.633 / 73.09
Qwen3-VL-2B, Euclidean	penalty, global mean cost	75.15 / 0.651 / 74.10	73.53 / 0.638 / 72.72
Qwen3-VL-2B, Euclidean	eligibility filter, global mean cost	75.15 / 0.651 / 74.10	73.53 / 0.638 / 72.72
Qwen3-VL-2B, Euclidean	penalty, state-conditional cost	75.15 / 0.651 / 74.10	73.53 / 0.638 / 72.72
MiniLM-L6-v2, text only	eligibility filter, global mean cost	75.55 / 0.656 / 74.43	74.19 / 0.653 / 73.24
MiniLM-L6-v2, text only	penalty, global mean cost	75.55 / 0.656 / 74.43	74.19 / 0.653 / 73.24
MiniLM-L6-v2, text only	eligibility filter, state-conditional cost	70.59 / 0.508 / 70.60	69.36 / 0.557 / 69.29

Table 9
Rank Score of the three Bench-CoE variants in accuracy-only and cost-aware forms, using TinyLLaVA for Query- and Subject-Bench-CoE. ΔR = 11-ID − 3-OOD, computed from the unrounded Rank Scores; bold marks the highest cost-aware Rank Score per view.

Variant	States K	Supervision read from the records	Cost term	11-ID	3-OOD	ΔR = 11-ID − 3-OOD
Query-Bench-CoE	N	per-sample labels	none	69.29	68.54	0.75
Query-Bench-CoE	N	per-sample labels	soft labels	73.83	63.32	10.52
Subject-Bench-CoE	30	subject leaderboard	none	69.73	51.00	18.73
Subject-Bench-CoE	30	subject leaderboard	penalty in the map	73.83	70.58	3.25
State-Bench-CoE	2	per-state profiles	none	72.42	69.93	2.49
State-Bench-CoE	2	per-state profiles	penalty in the map	**74.09**	**73.09**	1.00

7.9. Conclusions across the three variants

Table 9 and Fig. 10 collect the Rank Score of each variant in both forms, and three conclusions follow from them and from the subsections above.

1. Query-Bench-CoE improves its seen-dataset trade-off when the cost term is included, but the TinyLLaVA source-selected model reaches 73.83 on 11-ID, below the state-level router, and loses 10.52 Rank points on 3-OOD.
2. Subject-Bench-CoE needs no per-query records and holds the accuracy ceiling. Without a cost term it collapses under shift by 18.7 points, and with a cost term it loses 3.3.
3. State-Bench-CoE transfers best in either form. Without a cost term it is already the best accuracy-only router on the OOD datasets at 69.93, and with a cost term it loses 1.0 point under shift at less than a quarter of the strongest expert's cost.

Across the three variants, the cost term decides whether a router is economical and the granularity decides whether its economy survives a change of dataset.

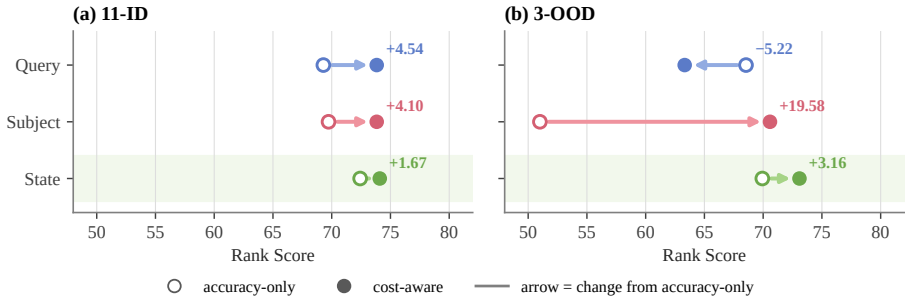


Figure 10: Rank Score of the three variants moving from accuracy-only to cost-aware forms on (a) 11-ID and (b) 3-OOD, with the change printed beside each row. TinyLLaVA is used at query and subject levels; the shaded row is the best cost-aware variant on the view.

8. Discussion and limitations

8.1. Which granularity, when

Because all three variants were built from the same records and compared under one protocol, their results guide the choice of granularity. Query-level routing with a cost term improves on its cost-free form when deployment resembles the source datasets, but the TinyLLaVA router scores 0.26 below State-Bench-CoE on 11-ID and 9.77 below it on 3-OOD. Subject-level routing is useful when a benchmark taxonomy matches deployment data: its cost-free form holds a seen-dataset accuracy ceiling, and adding cost to its map brings it within 0.26 Rank points of State-Bench-CoE on 11-ID and 2.51 on 3-OOD. State-level routing with a small K is strongest when transfer matters or the expert pool is cost-heterogeneous; the map-stability diagnostic selects K from source data. If two source halves disagree on a state's expert, the state is too small.

8.2. Accuracy ceiling or trade-off

Which operating point to prefer depends on the budget and on how far the deployment data may drift, and only the second consideration is settled here. The cost-free subject-level router keeps 81.2% in-distribution accuracy at 3.1 USD. The post-hoc hybrids keep 79.5–79.8% at 1.7–1.9 USD. Two-state State-Bench-CoE keeps 75.1% at 0.65 USD. The Rank Score with $\beta = 0.1$ encodes one preference between these, and other values of β move the optimum [cf. the β study of [4]]. What does not depend on β is the out-of-distribution finding: the cost-aware query-level router, which ties for second place in distribution, loses 10.52 Rank points under shift, and the two-state router loses about one.

8.3. Why not fine-tune a router

End-to-end routers and the state-level variant are complementary rather than competing. In the published OOD comparison RouterDC, VLC and MLP score 68.88, 69.44 and 55.79 on 3-OOD, below the 73.09 of two-state State-Bench-CoE. Local replays cover the other reference methods omitted from the OOD table of VL-RouterBench. Unlike

a recomputed Bench-CoE map, fine-tuned routers require retraining when the expert pool changes. A fine-tuned router could serve inside a state whose support is large, with the shared-state map as the fallback elsewhere, which is the structure the post-hoc study approximates. We leave a trained version of that design to future work.

8.4. What this says to dynamic classifier selection

The experiments answer a question that dynamic classifier selection raised but, under same-distribution evaluation, could not pose sharply: how large a competence region should be when the test distribution moves. State-Bench-CoE is clustering-and-selection [38] with three additions, and its answer, that under shift the variance of the competence estimate dominates and the region should be large, is consistent with the sensitivity to region size reported for local-accuracy selection [19] and with the motivation for pruning and local pools [44–46]. Those techniques were developed for pools of cheap base classifiers. The results here suggest they have a second life in routing among frozen foundation models.

8.5. More benchmarks, more supervision

The value of benchmarks for routing grows as more of them are released, because every new benchmark adds records that Bench-CoE turns into routing supervision without further inference or annotation. A benchmark that scores the current experts on new tasks extends the competence profiles to inputs the existing records do not cover. One that scores new experts lets the map include them by recomputation rather than retraining. One that is withheld from construction supplies a fresh OOD test of whether a router transfers. As benchmarks multiply, the same records serve more routers and more pools, the evidence behind each map grows while its cost stays at zero, and integrated benchmarks such as VL-RouterBench, which gather many datasets and experts under one protocol, become the natural place where this value accumulates.

8.6. Limitations

Three limitations remain.

1. The evidence comes from one benchmark. VL-RouterBench covers single-image prompts with binary, rule-based correctness and token-based price proxies, and some of its published reference-router figures do not reproduce from the released code and actions under a source-selected protocol. The framework is modality-agnostic, so a text-only replication on RouterBench [28] and graded competence profiles for open-ended outputs, to which the shrinkage estimator extends with a Dirichlet or Gaussian prior, are the natural next steps.
2. The expert pool and some settings are fixed. Routing to an expert without source records, as in UniRoute [17], requires profiling it on a calibration set before it enters the map. The OOD figures of the subject-level router carry the prototype caveat of Sec. 6.3.1; computed over MathVision and ChartQA alone, the two OOD

datasets that share no inputs with the prototypes, the cost-aware subject-level router scores 74.9 against 77.9 for State-Bench-CoE, so the ordering of Table 1 does not depend on the overlap. The post-hoc thresholds were tuned for the candidates of State-Bench-CoE rather than for each base router, which if anything disadvantages the hybrids.

3. The statistical guarantees are limited. The granularity analysis is a decomposition supported by diagnostics and sweeps rather than a finite-sample bound for the argmax map under covariate shift, and only State-Bench-CoE is reported over five seeds, while the cost-aware query- and subject-level routers and the retrained reference routers have been fitted once.

9. Conclusion

Building a router for a pool of frozen vision–language models does not require collecting routing supervision, because a benchmark that has scored every expert on every sample has already produced it. Bench-CoE builds every router from such records, writes it as a state assignment followed by a fixed state-to-expert map, and weighs each expert's cost alongside its accuracy, so that its three variants differ only in the number of states and can be trained and compared on one integrated benchmark under one protocol. On VL-RouterBench, with every operating point chosen on source data, the cost-aware query-level router improves its trade-off on the seen datasets but loses much of it on unseen ones, the subject-level variant holds the accuracy ceiling at a high cost and collapses under shift unless a cost term enters its map, and the state-level variant with a few learned states loses little under shift and transfers best at a small fraction of the strongest expert's cost. A matched-granularity comparison, a map-stability diagnostic, a post-hoc attribution study and an empirical-Bayes analysis agree on the reason: under dataset shift the variance of the competence estimate dominates its bias, so the object worth learning from benchmark records is a small set of shared, regularized and cost-aware competence profiles rather than a per-query rule. For practitioners the framework supplies routers that cost nothing to supervise, a map that is recomputed rather than retrained when the pool changes, and a label-free diagnostic for choosing how coarse that map should be.

CRediT authorship contribution statement

Yuanshuai Wang: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft. **Xingjian Zhang and Jinkun Zhao:** Data curation, Investigation. **Siwei Wen:** Formal analysis, Validation. **Shuhao Liao:** Investigation. **Xingchuang Liao:** Writing – original draft. **Junle Wang:** Resources. **Kui Zhang:** Formal analysis, Resources. **Lei Huang and Wenjun Wu:** Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The authors declare that no generative AI or AI-assisted technologies were used in the preparation of this manuscript.

Data availability

All datasets analysed or generated in this study are publicly available open-source benchmarks.

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